

Microbiological Analysis Report

Facility: Demo Foods Inc. | Product: Baby spinach | Date: 06/09/2026 | Analyst: Pedro Antonio Pérez Ferrer

Overall risk level: **CRITICAL**

Executive Summary

Testing of three baby spinach lots at Demo Foods Inc. revealed two non-conformances, including one critical finding: Salmonella spp. was detected in LOT-003, which constitutes an adulterant under FDA regulations and requires immediate lot rejection and hold. LOT-002 also showed elevated total coliform counts above acceptable limits, indicating a likely sanitation or cross-contamination issue in the production environment. Immediate corrective action is required to protect consumers, comply with FSMA regulations, and prevent further affected product from reaching commerce.

Analytical Results

Sample	Microorganism	Result	Limit	Status
LOT-001	Total coliforms	45 CFU/g	100 CFU/g	OK
LOT-001	Aerobic mesophiles	82,000 CFU/g	100000 CFU/g	OK
LOT-002	Total coliforms	320 CFU/g	100 CFU/g	WARNING
LOT-002	E. coli O157:H7	0 CFU/g	0 (absence) CFU/g	OK
LOT-003	Salmonella spp.	1 CFU/25g	0 (absence) CFU/25g	CRITICAL
LOT-003	Molds and yeasts	800 CFU/g	1000 CFU/g	OK

Required Corrective Actions

Sample LOT-001: All results for LOT-001 are within acceptable limits. Total coliforms at 45 CFU/g and aerobic mesophiles at 82,000 CFU/g are compliant, indicating this lot was produced under adequate sanitary conditions and has acceptable microbial quality for fresh baby spinach.

- No immediate corrective action required; continue routine monitoring per existing HACCP plan.
- Retain LOT-001 records as a baseline reference for acceptable microbial levels on this production line.

Sample LOT-002: LOT-002 has elevated total coliforms at 320 CFU/g, which exceeds the acceptable threshold and signals a breakdown in sanitation or a potential fecal contamination pathway somewhere in the production or post-harvest handling process. Although E. coli O157:H7 was not detected, elevated coliforms indicate increased risk of other pathogens being present.

- Place LOT-002 on immediate hold pending full investigation and additional confirmatory testing before any release decision.
- Conduct an immediate environmental swab program of all food-contact surfaces, wash tanks, conveyor belts, and drains on the affected production line.
- Verify sanitizer concentration, pH, and temperature in all wash water systems and document findings; recalibrate dosing equipment if out of specification.
- Review worker hygiene practices and GMP compliance at the time of LOT-002 production, including handwashing logs and glove change frequency.
- Retest LOT-002 with a statistically adequate number of samples before making a release or reject decision.
- Investigate irrigation water source and recent test results for indicator organisms.

Sample LOT-003: LOT-003 tested positive for *Salmonella* spp. at 1 CFU/25g. *Salmonella* is classified as a zero-tolerance adulterant pathogen in ready-to-eat leafy greens by the FDA — any detectable level is unacceptable and renders the product adulterated under the Federal Food, Drug, and Cosmetic Act. This is a critical food safety failure requiring immediate action. Mold and yeast counts are within acceptable range, suggesting the *Salmonella* contamination is likely of fecal or environmental origin rather than a broader spoilage issue.

- IMMEDIATELY reject and quarantine all units of LOT-003; do not release, ship, or use in any further processing.
- Initiate a mandatory FSMA Preventive Controls recall assessment and notify quality and regulatory leadership within the hour.
- If any LOT-003 product has already entered distribution, initiate a voluntary recall or market withdrawal in coordination with FDA and notify all affected customers and distributors.
- Shut down the implicated production line for thorough deep cleaning, sanitization, and environmental swabbing before resuming operations.
- Collect and test a minimum of 60 environmental swabs from food-contact and non-food-contact surfaces, drains, and high-risk zones to identify the environmental reservoir.
- Submit irrigation water, flume wash water, and final rinse water for immediate *Salmonella* testing.
- Review supplier certificates of analysis and conduct a traceback investigation to the farm of origin; consider suspending the implicated supplier pending investigation.
- Notify the farm supplier and conduct a joint root cause investigation per FSMA Produce Safety Rule requirements.
- Document all findings, actions, and timelines in the FSMA preventive controls corrective action log.

Regulatory Observations (FSMA)

- FSMA Preventive Controls for Human Food (21 CFR Part 117): Detection of *Salmonella* spp. in LOT-003 constitutes a hazard requiring a preventive control under the Hazard Analysis. Demo Foods Inc. must document a formal corrective action, conduct a root cause analysis, and evaluate whether a recall or withdrawal is necessary. Failure to act constitutes a violation of 21 CFR 117.150.
- FSMA Produce Safety Rule (21 CFR Part 112): Elevated total coliforms in LOT-002 and the *Salmonella* finding in LOT-003 may indicate non-compliance with agricultural water quality standards (Subpart E) and/or sanitation standards (Subpart L). The facility must conduct and document an investigation and implement corrective measures.
- FDA Adulteration under FD&C; Act Section 402(a)(1): LOT-003 containing detectable *Salmonella* spp. is deemed adulterated and cannot lawfully be sold or distributed in interstate commerce. Any distribution of this product prior to this finding must be assessed for potential recall notification to FDA under 21 CFR Part 7.
- FSMA Preventive Controls — Supply Chain Program (21 CFR Part 117 Subpart G): If *Salmonella* contamination is traced to a raw material supplier or farm of origin, Demo Foods Inc. must have and apply a documented supply chain program, including supplier verification activities such as onsite audits or testing, and must reassess the supplier's approved status.
- FSMA Records Requirements (21 CFR Part 117.190): All corrective actions, environmental monitoring results, hold decisions, and retest data must be documented, signed, and retained for a minimum of 2 years and must be available for FDA inspection upon request.

Critical Control Point Recommendations

- CCP-1 — Wash Water Antimicrobial Treatment: Establish a validated CCP for sanitizer concentration (e.g., chlorine 50–200 ppm free chlorine, pH 6.0–7.0, or equivalent validated alternative) in all flume wash and spray wash systems, with continuous or real-time monitoring, documented at minimum every 30 minutes per shift, and defined critical limits with immediate corrective action procedures upon deviation.
- CCP-2 — Incoming Water Source Verification: Establish a CCP or prerequisite program for irrigation and process water quality, requiring validated testing for generic *E. coli* and *Salmonella* at defined frequencies consistent with FSMA Produce Safety Rule (21 CFR Part 112) standards, with defined action thresholds and hold procedures.

- CCP-3 — Post-Harvest Environmental Pathogen Monitoring: Implement a robust environmental monitoring program (EMP) as a CCP supporting measure, with defined sampling zones (Zones 1–4), minimum sampling frequency (weekly for high-risk zones), and a documented trend analysis and corrective action protocol triggered by any positive Salmonella finding.
- CCP-4 — Supplier Raw Material Verification: Establish a supplier verification CCP requiring receipt of a Certificate of Analysis (COA) with Salmonella and indicator organism testing for each incoming lot, with a documented hold-and-test procedure before any lot enters processing when COA is absent, out-of-spec, or from a new/unapproved supplier.
- CCP-5 — Temperature Control Throughout Processing: Verify and document product temperature at receiving, during processing, and at cold storage entry does not exceed 4°C (41°F), as temperature abuse is a contributing factor to pathogen proliferation in fresh leafy greens; establish critical limits with calibrated monitoring devices and corrective action procedures.

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